

CTE-log monitoring and analysis on window systems

Window logs are configured, log retention, log size based on the registry key

HKEY LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\EventLog\<Event Log>

Below there's many more sub keys which are CustomID, DisplaynameID, File, MaxSize, etc.

The OS database data are stored in the System.evtx and security in Security.evtx, application in Application.evtx, All of .evtx file can be open and read with event logs all a stored in

C:\Windows\System32\winevt\Logs

Log Entries

- Application event log
 - Logged by the application and program like installing packages, denying AV or shutdown firewall and such
- Security event log
 - Security log is the mother of all logs in forensic term but turn off by default (eg, log ons/off, attempted connections, policy changes
- Setup event log
- System event log
 - It contain the logs which happen around in the OS such as
 - Changes to OS
 - Hardware configuration
 - Driver installation
 - Start/Stop services
 - Log of failures ...
- Forwarded event log
- Custom log

Event Log Types

- Error
- Warning
- Information
- Successful Audit
- Failure Audit

Event Log Panel Information

- Log Name
- Source
- EventID
- Level
- User
- Logged
- Task Category
- Computer

1. The "Control Room" (Registry & Storage)

Before any logs are written, the system needs to know the rules.

- **The Brain (Registry):** Located at `HKEY_LOCAL_MACHINE...\EventLog`. This tells Windows how big a log file can get (`MaxSize`) and where to save it (`File`). If the log gets too big, the registry determines if it should overwrite old data or stop recording.
- **The Cabinet (Files):** All logs are saved as `.evtx` files in `C:\Windows\System32\winevt\Logs`. You can manually copy these files to another computer to analyze them later—this is a huge part of **Digital Forensics**.

2. The "Logbooks" (Main Categories)

Windows doesn't put everything in one pile; it uses specific "books" for different activities:

- **System Log:** The "Hardware & OS" book. It tracks driver installs, Windows updates, and if your power supply is failing.
- **Security Log:** The "**Mother of All Logs.**" It tracks who came in and out (Logons). **Crucial note:** It's mostly silent by default; you have to turn on "Audit Policies" to make it start recording.
- **Application Log:** The "Software" book. If Chrome crashes or your Antivirus blocks a file, it goes here.
- **Setup Log:** Specifically for Windows installations and major updates.
- **Forwarded Logs:** Used in offices where one "Master" computer collects logs from all the other computers on the network.

3. The "Urgency Level" (Event Types)

When you look at a log, the **Level** tells you how much you should panic:

Level	Meaning
Information	"Everything is fine, just letting you know I started a task."
Warning	"Something is weird. It's not broken yet, but keep an eye on it."
Error	"Something failed. A service didn't start or an app crashed."
Success Audit	"A user tried to access something and was allowed ."
Failure Audit	"A user tried to access something and was blocked ."

4. Reading the Entry (The Panel Info)

Every single log entry contains specific "metadata" so you can investigate it:

- **EventID:** A unique number code (e.g., **4624** is a successful logon). **Pros use this to Google solutions quickly.**
- **Source:** Which specific program or "sub-system" created the log.
- **User:** Which account was active when this happened.
- **Logged:** The exact date and time (vital for building a timeline of an attack).
- **Task Category:** A label that helps group similar types of events together.